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Confidence-guided dual hypergraph learning for sparse-label graph classification

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ABSTRACT

Graph classification is crucial for analyzing various complex relational patterns, but existing methods heavily rely on large amounts of labeled data, the acquisition of which is often costly or infeasible. Although Semi-Supervised Learning (SSL) partially alleviates the issue of data scarcity, current approaches still face two major limitations: (1) Graph Neural Networks (GNNs) struggle to effectively model high-order semantic relationships; (2) The uniform treatment of unlabeled data introduces a confidence crisis, which leads to performance degradation. To address these challenges, we propose a novel semi-supervised graph classification framework—**CGD-Hyper**, which integrates a dual hypergraph fusion mechanism and a reliability-aware label encoding strategy. The core innovations of this framework lie in: firstly, designing a dual hypergraph fusion mechanism that jointly encodes graph structural topology and node features to explicitly and effectively model high-order semantic interactions; secondly, introducing a reliability-aware method that dynamically filters high-confidence unlabeled samples and generates robust supervisory signals for training through mixed label encoding. Experiments on four benchmark datasets demonstrate the superior performance of **CGD-Hyper**, achieving an average accuracy improvement of 3.5% over the state-of-the-art methods.

1. Introduction

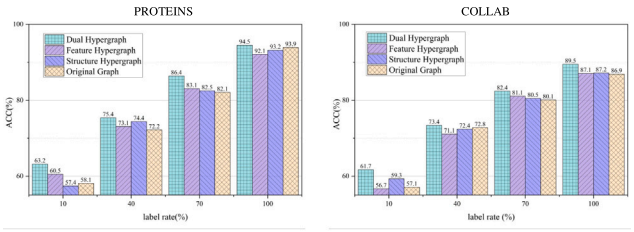
As a paradigm for structured data representation, graphs provide a powerful framework [1,2] for modeling complex systems, achieved by mapping entities to nodes and relationships to edges. Their inherent strength [3] lies in their ability to simultaneously capture a system's topological structure and the semantic associations between entities. To fully exploit the latent relational patterns and system-level insights embedded within graph-structured data, graph classification has emerged as a critical research direction. This task aims to assign a category label to an entire graph instance by learning hierarchical semantic representations of graphs, which in turn supports practical applications such as detecting anomalous communities in social networks or predicting molecular functional properties.

In recent years, GNNs [4,5] have achieved significant progress and garnered widespread attention in graph classification tasks, largely due to their unique ability to perform end-to-end graph embedding learning through iterative neighbor aggregation. Nevertheless, conventional GNN methods are primarily limited to modeling local, low-order relationships [6] (e.g., direct neighbors) and often overlook high-order semantic patterns (e.g., complex interactions among multiple nodes) that hold global significance. To overcome this limitation, Hypergraph Neural Networks [7] (HGNNs) have been introduced by utilizing hyperedges to explicitly connect multiple nodes. Despite these advancements, a fundamental challenge persists: these methods generally rely heavily on large-scale, high-quality labeled datasets. However, acquiring such labels is often not only prohibitively expensive but also potentially infeasible in many real world application scenarios, constituting a major barrier to the widespread deployment of graph classification models.

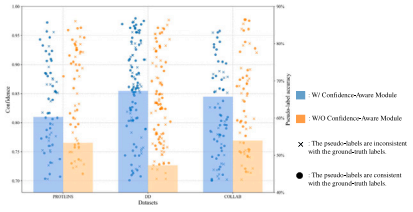
To address the scarcity of labeled data, strategies such as pre-training [8], active learning [9], and semi-supervised learning [10] have been extensively explored. Nevertheless, applying these paradigms to hypergraph structures presents ongoing challenges: firstly, the reliance of existing hypergraph methods [11] on a single hypergraph type limits their capacity to represent the diverse high-order relationships encountered in the real world; secondly, existing frameworks [12] often lack effective mechanisms for differentiating unlabeled data, tending instead to utilize all available data indiscriminately. This approach is prone to introducing noise from low-quality

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(a) Performance comparison on PROTEINS (left) and COLLAB (right) datasets under different label ratios and models.



(b) Pseudo-label confidence distribution and accuracy comparison before vs. after applying the dual confidence-aware strategy on PROTEINS, DD and COLLAB.

Fig. 1. An illustration of two key limitations in traditional methods: (a) the restriction of modeling with single-type hypergraphs, and (b) the noise problem introduced by low-quality pseudo-labels.

pseudo-labels, which can subsequently affect the training stability and generalization performance of the model.

To elucidate the limitations of single hypergraph modeling and the noise problem caused by low-quality pseudo-labels, we conduct experimental analyses. First, on two standard semi-supervised graph classification datasets (statistical details are presented in Section 5), we perform comparative experiments using labeling ratios of 10%, 40%, 70%, and 100%. As shown in Fig. 1(a), we compare the performance of models using dual hypergraphs, feature hypergraphs, structural hypergraphs, and original graphs as inputs (all models are trained using standard cross-entropy loss). The experimental results indicate that as the labeling ratio decreases (i.e., weak supervision strengthens), single hypergraph models generally achieve lower performance than original graph models. For instance, on the PROTEINS dataset, when the labeling ratio is as low as 10%, the structural hypergraph model achieves an accuracy of only 57.4%, whereas the original graph model reaches 58.1%. This highlights the difficulty of single hypergraph models in effectively capturing the diverse high-order relationships within graphs, leading to suboptimal learned representations. In contrast, the dual hypergraph model significantly outperforms all single hypergraph models and the original graph model across all datasets and labeling ratios, validating that the dual hypergraph fusion mechanism can capture high-order semantic structures more comprehensively and effectively compensate for the limitations of single hypergraph modeling.

Subsequently, we further evaluate the accuracy of pseudo-labels before and after the introduction of the confidence-aware module on three datasets. As illustrated in Fig. 1(b), the basic pseudo-label generation method (without confidence filtering) shows generally low accuracy (approximately 55%), even falling below 50% on the DD dataset, indicating significant unreliability. Analysis of the confidence distribution reveals that traditional methods still exhibit a high error rate within the high-confidence interval (0.9–1.0), resulting in a confidence crisis. Direct use of these pseudo-labels would introduce substantial noise, impairing the stability of model training and its generalization capability. In contrast, the confidence-aware module significantly enhances the reliability of pseudo-labels, particularly alleviating the confidence crisis in the high-confidence region by dynamically filtering high-confidence samples, thereby substantially reducing noise interference.

Building upon our preceding analyses and findings, We propose a novel semi-supervised framework, CGD-Hyper (Confidence-Guided Dual Hypergraph Learning for Sparse-Label Graph Classification), comprising two synergistic modules: (1) The Dual Hypergraph Representation Module constructs structural and feature hypergraphs to model topological high-order relationships and feature-based semantic interactions, employs parameter-shared adaptive encoders to learn correlated node-hyperedge embeddings, and fuses them via collaborative gating with learnable attention weights supervised by hyperedge homogeneity loss; and (2) The Confidence-Aware Optimization Module implements a dual-confidence assessment combining clustering and classification to identify reliable unlabeled graphs for mixed label encoding, then jointly optimizes supervised loss from labeled graphs and pseudo-label encoding loss from high-confidence unlabeled graphs through end-to-end training. This integrated design mitigates pseudo-label noise while enhancing high-order relationship modeling under low-labeling regimes. Specifically, the contributions of this paper are as follows:

- **Dual-Hypergraph Fusion Mechanism**: We design a dual-hypergraph fusion framework to simultaneously capture structural dependencies and feature-based semantic associations within graph data by constructing structural and feature hypergraphs and jointly learning these two types of information, thereby effectively uncovering diverse high-order interactions that are often inaccessible to traditional single-hypergraph models.
- **Confidence-Aware Label Encoding**: To address the confidence crisis in semi-supervised learning, we propose a confidence-aware label encoding strategy, which combines classification prediction with clustering analysis to assess the prediction confidence of unlabeled graph instances, thereby filtering out high-confidence samples.
- **Experimental Validation**: We conduct a comprehensive evaluation of CGD-Hyper on four datasets covering three kinds of domains. The experimental results show that this method achieves an average improvement of 3.5% in graph classification accuracy, surpassing the latest baseline models. Notably, it demonstrates significant performance advantages, especially under conditions of low labeling ratios.

2. Related work

2.1. Hypergraph learning

In recent years, hypergraphs have gained significant attention [13] due to their ability to model complex high-order relationships. Unlike ordinary graphs, a hyperedge [14] in a hypergraph can connect multiple nodes simultaneously, enabling more flexible capture of high-order dependencies among nodes. Leveraging this property, researchers [15] have proposed methods to construct hypergraphs and extract semantic information using hypergraph neural networks, which can be categorized into two types: single-hypergraph construction and hypergraph-line graph joint construction. In single-hypergraph construction, models typically utilize a single type of relationship (e.g., feature similarity or adjacency) to build hypergraphs. For example, HDS [16] constructed dynamic feature hypergraphs based on node feature similarity to extract high-order feature information, while HGNN [17] built hyperedges using node adjacency relationships to capture high-order structural information. In hypergraph-line graph joint construction, HEAL [18] designed a learnable hypergraph structure, avoiding predefined distance- or feature-based criteria in traditional hypergraph construction, and combined it with line graphs to extract relationships among high-order substructures.

Despite their improvements, these methods still face limitations: existing approaches often rely on a single type of relationship to construct hypergraphs, failing to comprehensively capture diverse high-order semantic information within graphs.

2.2. Semi-supervised learning

Semi-supervised learning [19] leverages a small amount of labeled data and a large amount of unlabeled data to provide richer supervision signals for model training. Existing methods [20] can be broadly categorized into knowledge distillation-based and contrastive learning-based approaches. In knowledge distillation-based methods, student models acquire additional supervision by mimicking the knowledge of teacher models. For example, NoisyStudent [21] introduced noisy training into the knowledge distillation framework to enhance the robustness of student models under label scarcity. In contrastive learning-based methods, researchers generate perturbed views of samples to encourage consistency among different views of the same sample in the feature space. For instance, InfoGraph [22] and GraphCL [23] applied graph contrastive learning to semi-supervised classification tasks, significantly improving model performance. Additionally, TGNN [24] effectively leverages unlabeled data by exploiting the similarity between probability distributions derived from message-passing mechanisms and graph kernel functions. In a different vein, HEAL [18] innovatively utilized hypergraph neural networks to extract high-order semantic information within graphs, further advancing semi-supervised learning. Furthermore, recent advancements explore disentangled representation learning for enhanced robustness. For instance, DisenSemi [25] maximizes the mutual information between factual graphs and their corresponding representations while minimizing the redundancy among representations derived from distinct factual graphs.

Despite these achievements, semi-supervised learning methods still suffer from the following limitations: (1) existing approaches primarily rely on GNNs to capture low-order neighborhood information, failing to fully exploit complex high-order semantic relationships within graphs; (2) most methods lack selectivity when processing unlabeled data, potentially introducing noisy samples that degrade model performance.

2.3. Pre-training learning

Pre-training [26] enhances model generalization on downstream tasks by enabling representation learning from large-scale unlabeled data. Pre-training approaches can be categorized into autoencoder-based, generative model-based, and contrastive learning-based methods [27]. Autoencoder-based methods employ encoder–decoder architectures to reconstruct input signals, encouraging the encoder to extract salient semantic features. For instance, Surface-VQMAE [28] introduces vector quantization enhanced masked autoencoding for learning discrete latent representations of surface point clouds. Generative model-based methods, such as GPT-GNN [29], perform graph generation tasks to jointly model node attributes and structural dependencies, optimizing the likelihood of the entire attributed graph. Contrastive learning-based methods generate augmented graph views and optimize representation consistency across positive pairs. For example, MoICLR [30] constructs molecular graph variants and uses a contrastive estimator to learn robust chemical representations.

However, existing pre-training methods face two main challenges. First, most adopt a two-stage training pipeline, which separates the utilization of labeled and unlabeled data and hinders joint optimization. Second, they predominantly rely on conventional graphs for structural encoding, failing to explore the representational benefits of hypergraphs in capturing high-order dependencies during pre-training.

2.4. Active learning

Active learning aims [31] to select the most informative samples from unlabeled graphs for annotation under a limited labeling budget. Based on sample selection strategies, existing methods [32] can be divided into single-objective and multi-objective approaches. In single-objective methods, researchers [33] focus on sample diversity or model performance changes. For instance, SMQP [34] proposed a diversity-based sample selection criterion by combining clustering and fuzzy rough set theory to maximize input space coverage, while MCDAL [35] employed two auxiliary classification layers to learn stricter decision boundaries by maximizing the discrepancy between them, and utilized the differences among classifiers to acquire the function for active learning. In multi-objective methods, GraphSpa [12] simultaneously considered local similarity and global semantic structures, integrating contrastive learning to combine semi-supervised learning with active learning, significantly improving sample selection effectiveness.

However, existing active learning methods still face challenges when dealing with large-scale unlabeled graphs: although selecting high-quality graphs for annotation alleviates label scarcity, the overall utilization of unlabeled graphs remains low, limiting the model's ability to simulate the input data distribution.

3. Preliminaries

Hypergraph. A hypergraph is a generalization of a simple graph, where a hyperedge can connect more than two nodes. Formally, a hypergraph is represented as $G_H = (V, E_H)$, where V is a set of n nodes and E_H is a set of hyperedges. Each hyperedge is a subset of V, which degenerates into a simple edge connecting only two nodes in a simple graph. Let $\mathbf{X} \in \mathbb{R}^{n\times d}$ denote the d-dimensional attributes of the n nodes, and $\mathbf{H} \in \mathbb{R}^{n\times m}$ represent the incidence matrix of the hypergraph, where $\mathbf{H}_{ij} = 1$ indicates that node i belongs to the jth hyperedge, and m is the number of hyperedges. For G_H corresponding to a simple graph G, its label is denoted as $\mathbf{y} \in \mathbb{R}^{C\times 1}$, which is a one-hot encoding vector where C is the number of classes.

Hyperedge Homogeneity. To quantify the inter-dependencies among nodes within a hyperedge, we introduce hyperedge homogeneity as a metric. We hypothesize that the stronger the correlation between nodes within a hyperedge, the higher the homogeneity of that hyperedge, and consequently, the greater its significance in the

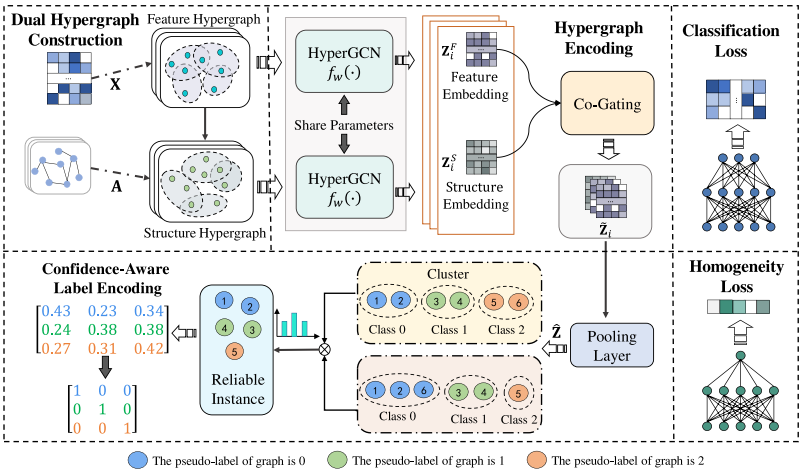


Fig. 2. The framework of CGD-Hyper.

message-passing mechanism. Specifically, consider a hypergraph $G_H = (V, E_H)$, where the egonet $E(v_i)$ of a node $v_i \in V$ is defined as the set of all hyperedges containing v_i. We employ the number of shared hyperedges between nodes as a measure to assess the degree of correlation between them. Ultimately, the average correlation degree among all pairs of nodes within a hyperedge is defined as the homogeneity score of that hyperedge, as shown in the following formula:

$$homo(e) := \begin{cases} \sigma\left(\frac{\sum_{\{v_i,v_j\}\in\binom{e}{2}} |E(v_i)\cap E(v_j)|}{|\binom{e}{2}|}\right) & \text{if } |e|>1, \\ 1 & \text{otherwise,} \end{cases} \tag{1}$$

where $\binom{e}{2} = \{(v_i, v_j) \mid v_i, v_j \in e \text{ and } i < j\}$ represents all node pairs within the hyperedge e, $|\binom{e}{2}|$ is the cardinality of $\binom{e}{2}$, and $|E(v_i) \cap E(v_j)|$ denotes the number of common hyperedges between nodes v_i and v_j. $\sigma(\cdot)$ represents sigmoid function.

Semi-supervised Graph Classification. Given a graph set $\mathcal{G} = \{G_1, G_2, \ldots, G_N\}$ divided into two disjoint subsets: a labeled graph set $\mathcal{G}^L$ and an unlabeled graph set $\mathcal{G}^U$ (where $\mathcal{G}^L \cap \mathcal{G}^U = \emptyset$ and $\mathcal{G}^L \cup \mathcal{G}^U = \mathcal{G}$). The goal is to learn a predictive function $f : G \to \mathbf{y}$ that generalizes from $\mathcal{G}^L$ to accurately infer labels for the graphs in $\mathcal{G}^U$, leveraging both the labeled and unlabeled graphs for improved learning.

4. Methodology

In this section, we introduce the CGD-Hyper framework, which is designed to tackle the problem of label scarcity in graph classification tasks. As illustrated in Fig. 2, firstly, to effectively capture diverse high-order semantic relationships in graph data, we construct hypergraphs from both feature and structural perspectives using a dual hypergraph construction module. Secondly, considering the potential correlations between nodes in hypergraphs, we employ two adaptive hypergraph encoders with shared parameters [17,36] to encode the hypergraphs from these two perspectives, and their representations are fused through a collaborative gating mechanism. Finally, in addition to providing supervision signals by applying cross-entropy loss to the limited labeled graphs, we introduce a confidence-aware label encoding method to filter out high-quality graph samples for label encoding, thereby further optimizing the model training process.

4.1. Dual-hypergraph construction

To overcome the limitation of traditional hypergraph methods that rely solely on a single hypergraph representation, we propose an innovative dual-hypergraph framework. This framework facilitates the effective propagation of node features across both topological and feature spaces. By constructing two complementary hypergraphs – one based on node attributes forming a feature hypergraph, and the other based on adjacency relationships forming a structural hypergraph – our approach enables the in-depth capture of complex semantic patterns, thereby achieving a richer and more refined representation of intricate interrelationships.

4.1.1. Feature hypergraph construction

We construct a feature hypergraph G_H^F by connecting each node with its top-k similar nodes to form feature hyperedges. Specifically, we employ the cosine similarity function to measure the feature correlation between nodes, inherently yielding values in $[-1, 1]$ and obviating the need for additional normalization. We denote the resulting similarity value between nodes v_i and v_j as $\mathbf{S}_{ij}^F$, which forms the similarity matrix $\mathbf{S}^F \in \mathbb{R}^{n\times n}$.

In the feature hypergraph construction process, each node represents an entity, and each hyperedge is formed by the entity and its top-k similar neighbors. The construction of feature hypergraph can be defined as:

$$\mathbf{H}_{ij}^F = \begin{cases} 1, & \text{if } j \in T(i) \text{ or } i = j, \\ 0, & \text{else,} \end{cases} \tag{2}$$

where $\mathbf{H}^F \in \mathbb{R}^{n\times n}$ is the incidence matrix of the feature hypergraph G_H^F, and $T(\cdot)$ is a selection function that search the top-k most similar nodes based on the similarity matrix $\mathbf{S}^F$.

4.1.2. Structure hypergraph construction

In this subsection, we introduce a structure hypergraph G_H^S designed to complement the feature hypergraph. Specifically, in the structure hypergraph, each node corresponds to a entity, and every hyperedge connects the entity to its local ego-network neighbors as follows:

$$\mathbf{H}_{ij}^S = \begin{cases} 1, & \text{if } j \in \mathcal{N}(i) \text{ or } i = j, \\ 0, & \text{else,} \end{cases} \tag{3}$$

where $\mathbf{H}^S \in \mathbb{R}^{n\times n}$ denotes the incidence matrix of the structure hypergraph G_H^S, and $\mathcal{N}(i)$ represents the set of neighbors connected to node i.

Note that real-world graph data frequently includes isolated or low-degree nodes, which suffer from inadequate connectivity during the construction of the structure hypergraph, thereby hindering efficient message propagation. To mitigate this, we introduce a hyperedge size threshold α. If a hyperedge's size (i.e., the number of nodes it connects) is below α, we systematically incorporate feature-derived hyperedges from the feature hypergraph into the structure hypergraph as follows:

$$\mathbf{H}_j^S = \begin{cases} \mathbf{H}_j^F, & \text{if } \sum_{i=1}^n \mathbf{H}_{ij}^S < \alpha, \\ \mathbf{H}_j^S, & \text{else.} \end{cases} \tag{4}$$

4.2. Hypergraph encoding

Based on the aforementioned methods, we have successfully constructed both structural hypergraphs and feature hypergraphs. However, how to effectively extract high-order features from hypergraphs remains a critical challenge. Conventional hypergraph neural networks (e.g., HGNN [17]) establish message-passing channels based on the affiliation between hypernodes and hyperedges, yet this approach fails to adequately model latent dependencies among nodes, resulting in suboptimal representation learning. Inspired by dynamic hypergraph neural networks (CHGNN [37]), we propose an adaptive hypergraph encoder that: (1) leverages hyperedge homogeneity to characterize latent node relationships and integrates them into the message-passing process; (2) employs a co-gating mechanism to dynamically fuse dual hypergraph representations. The detailed design is presented below.

4.2.1. Hypergraph encoder

To effectively utilize the extracted high-order feature and structural information within the graph, we employ two adaptive hypergraph neural networks with shared parameters, $f_w(\cdot)$ and $f_s(\cdot)$, to encode the feature hypergraph and structure hypergraph, respectively. Specifically, the feature hypergraph $f_w(\cdot)$ encoding is defined by:

$$(\mathbf{Z}^F)^{l+1} = \sigma\left((\mathbf{D}_v^F)^{-\frac{1}{2}}\mathbf{H}^F homo(G_H^F)(\mathbf{D}_e^F)^{-1}(\mathbf{H}^F)^T(\mathbf{D}_v^F)^{-\frac{1}{2}}(\mathbf{Z}^F)^l\mathbf{\Theta}\right), \tag{5}$$

where $(\mathbf{Z}^F)^{l+1} \in \mathbb{R}^{n\times d_1}$ represents the feature hypergraph encoding at the $(l+1)$th layer, $(\mathbf{Z}^F)^0 = \mathbf{Z}$. Given an feature hypergraph incidence matrix $\mathbf{H}^F$, a node $v \in V$, and a hyperedge $e \in E_H$, we define the degree of node v as $d(v) = \sum_{e\in E_H}\mathbf{H}_{ve}^F$, and the degree of hyperedge e as $d(e) = \sum_{v\in V}\mathbf{H}_{ve}^F$. Besides, $\mathbf{D}_v^F$ and $\mathbf{D}_e^F$ denote the diagonal matrices of node degrees and hyperedge degrees in the feature hypergraph. $homo(G_H^F) \in \mathbb{R}^{m\times m}$ represents a diagonal matrix composed of the homogeneity of all hyperedges in feature hypergraph G_H^F. $\mathbf{\Theta}$ is a trainable parameter matrix.

Similarly, to extract high-order information $\mathbf{Z}^S$ from the structural hypergraph, we utilize a hypergraph neural network $f_s(\cdot)$, which shares parameters with $f_w(\cdot)$, for the purpose of structural encoding.

4.2.2. Co-gating mechanism

With the graph representations from the feature hypergraph perspective ($\mathbf{Z}^F$) and the structure hypergraph perspective ($\mathbf{Z}^S$) obtained, we aim to integrate them in a manner that enables mutual supervision and enhancement. While prior work has explored fusion methods such as attention mechanisms, concatenation with MLP projection, and direct summation, these approaches often fail to explicitly model the interplay between the two perspectives. Inspired by [38], we propose a Co-Gating (CG) to dynamically balance and integrate $\mathbf{Z}^F$ and $\mathbf{Z}^S$, thereby preserving their complementary information while allowing mutual correction. The fusion process is formalized as:

$$\tilde{\mathbf{Z}} = \mathbf{B}\odot(\mathbf{Z}^F)^L + (1-\mathbf{B})\odot(\mathbf{Z}^S)^L, \tag{6}$$

where $\tilde{\mathbf{Z}} \in \mathbb{R}^{n\times d_1}$ denotes the fused representation matrix, $\mathbf{B} \in \mathbb{R}^{n\times d_1}$ is the corresponding element-wise fusion weight matrix, and $\odot$ indicates the Hadamard product. The weight matrix $\mathbf{B}$ is dynamically computed through a gating mechanism as defined in Eq. (7):

$$\mathbf{B} = \sigma\left(\mathbf{W}_g\cdot[(\mathbf{Z}^F)^L,(\mathbf{Z}^S)^L] + \mathbf{b}_g\right), \tag{7}$$

where $\mathbf{W}_g \in \mathbb{R}^{n\times 2n}$ is a learnable weight matrix, and $\mathbf{b}_g \in \mathbb{R}^{n\times d_1}$ is the bias term. The notation $[\cdot,\cdot]$ represents column-wise concatenation of two matrices.

Subsequently, we employ an average pooling function to extract the graph-level representation for each graph. In detail, the process for graph-level representation of the ith graph $\hat{\mathbf{Z}}_i$ is delineated by:

$$\hat{\mathbf{Z}}_i = \frac{1}{n}\sum_{j=1}^{n}\tilde{\mathbf{Z}}_j^i. \tag{8}$$

4.3. Confidence-aware label encoding

To effectively utilize unlabeled graph data and circumvent the confidence crisis arising from sample unreliability, we propose a Confidence-Aware Label Encoding mechanism. This method integrates confidence scores from both clustering and classification dimensions to screen highly reliable samples, based on which label encoding is performed to provide supplementary supervision.

4.3.1. Confidence-aware reliable graph extraction

To select reliably labeled unlabeled graphs, we design a confidence alignment mechanism. The core of this mechanism lies in leveraging the consistency between the confidence scores derived from clustering and classification tasks for unlabeled graphs, thereby identifying and selecting high-reliability samples. Specifically, we employ a clustering algorithm $f_{\text{cluster}}(\cdot)$ to cluster the learned embeddings of unlabeled graphs into C' clusters. Typically, the number of clusters C' exceeds the true number of classes C in the dataset. This clustering process can be formally defined as:

$$f_{cluster}(\hat{\mathbf{Z}}^U) \longrightarrow \mathcal{K}'_1, \mathcal{K}'_2, \ldots, \mathcal{K}'_{C'}, \tag{9}$$

where $\hat{\mathbf{Z}}^U \in \mathbb{R}^{|\mathcal{G}^U|\times d_1}$ is the representations of unlabeled graphs, and $\mathcal{K}'_{C'}$ represents the C'th cluster of unlabeled graph. Besides, we divide $\mathcal{G}^L$ into C clusters $\{\mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_C\}$ with true labels.

Then, we compute the embedding centers $\mathbf{q}_c^L, \mathbf{q}_{c'}^U \in \mathbb{R}^{d_1}$ for each category in both labeled and unlabeled graphs as follows:

$$\mathbf{q}_c^L = \text{average}(\hat{\mathbf{Z}}_j^L | G_j \in \mathcal{K}_c) | c \in \{1,2,\ldots,C\}, \tag{10}$$

$$\mathbf{q}_{c'}^U = \text{average}(\hat{\mathbf{Z}}_j^U | G_j \in \mathcal{K}'_{c'}) | c' \in \{1,2,\ldots,C'\}, \tag{11}$$

where average$(\cdot)$ function represents the mean operation.

Subsequently, the label information from labeled graphs is transferred to unlabeled graphs through nearest centers matching $\tilde{\mathcal{G}}_c^U = \bigcup_{\mathbb{I}\left(\arg\min_c D(\mathbf{q}_c^L, \mathbf{q}_{c'}^U)\right)}\mathcal{K}'_{c'}$, where $\mathbb{I}\left(\arg\min_c D(\mathbf{q}_c^L, \mathbf{q}_{c'}^U)\right)$ is the indicator function, $D\left(\mathbf{q}_c^L, \mathbf{q}_{c'}^U\right)$ is a Euclidean distance function.

Building upon the aforementioned method, we utilize clustering techniques to partition the unlabeled data into C categories. To quantify the category assignment confidence for each unlabeled data point within the clustering results, we adopt cosine similarity as the metric. Specifically, for a graph G_i^U, its confidence $\mathbf{y}_{ic}$ in category c is defined as the cosine similarity between the graph and the corresponding class center $\tilde{\mathbf{q}}_c^U$. The calculation is given by:

$$\mathbf{y}_{ic} = simi\left(\hat{\mathbf{Z}}_i^U, \tilde{\mathbf{q}}_{c'}^U\right), \tag{12}$$

where $simi(\cdot)$ represents a function for calculating similarity. Based on the above method, we obtain clustering-based confidences $\mathbf{y}_i \in \mathbb{R}^C$ for unlabeled graphs. To enhance reliability, we further incorporate classification-based confidences $\hat{\mathbf{y}}_i \in \mathbb{R}^C$ generated by the classifier $f(\cdot)$. We then compute the Hadamard product between $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ to derive a more reliable confidence estimate: $y_i^* = argmax(\mathbf{y}_i \odot \hat{\mathbf{y}}_i)$, where $argmax(\cdot)$ selects the maximum value as the final confidence.

Therefore, leveraging the inter-supervision mechanism between clustering and classification, we obtain a reliable confidence estimate regarding the category assignment of unlabeled data. Specifically, the Hadamard product acts as a consistency check: when the two confidences conflict, samples are deemed unreliable; conversely, they are confirmed as reliable. This mechanism effectively filters out noisy supervision signals. Furthermore, to mitigate the potential risk of underfitting during model training, we discard the hard-threshold strategy that exclusively selects high-confidence samples. Instead, we treat the fused confidence y_i^* as a probability distribution and perform probabilistic sampling based on it to select graphs for label encoding. Ultimately, the resulting reliable sample set obtained via this strategy is denoted as $\hat{\mathcal{G}}^U$.

4.3.2. Label encoding

To fully leverage the hidden information contained in the unlabeled graphs for model enhancement, we introduce a label encoding mechanism [39] that constructs a prediction means matrix $\hat{\mathbf{M}}^U \in \mathbb{R}^{C\times C}$ from the predictions of reliable unlabeled graphs, as follows:

$$\hat{\mathbf{M}}_c^U = \frac{1}{\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U} \left(\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U \hat{\mathbf{y}}_i^U \right), \tag{13}$$

where $\hat{\mathbf{M}}_c^U$ denotes the label encoding for the cth class, while $\hat{\mathbf{y}}_i^U$ represents the classification outcome of the ith unlabeled graph. In Theorem 4.1, we theoretically derive and characterize the essential properties that $\hat{\mathbf{M}}_c^U$ must satisfy. A detailed proof of Theorem 4.1 is provided in Appendix.

Theorem 4.1. *$\hat{\mathbf{M}}_c^U$ satisfies the following properties:*

- *Let $\mathbf{1} \in \mathbb{R}^{|C|}$ be an all-ones vector, there exists $\mathbf{1}^T \hat{\mathbf{M}}_c^U = 1$.*
- *$0 \le \hat{\mathbf{M}}_{c,j}^U \le 1, \forall j \in \{1, \dots, |C|\}$, where $\hat{\mathbf{M}}_{c,j}^U$ denotes the jth element of $\hat{\mathbf{M}}_c^U$.*
- *For any $i \in \{1, \dots, |\hat{\mathcal{G}}^U|\}$, if the classification prediction $\hat{\mathbf{y}}_i^U$ of graph G_i corresponds to its true label, then $\hat{\mathbf{M}}_c^U$ is equal to the true label encoding $\mathbf{M}_c$, where $\mathbf{M} \in \mathbb{R}^{C\times C}$ represents the ground-truth label encoding.*
- *If $\hat{\mathbf{M}}_c^U$ equals $\mathbf{M}_c$ for some $c \in \{1, \dots, |C|\}$, then for any $i \in \{1, \dots, |\hat{\mathcal{G}}^U|\}$, $\hat{\mathbf{y}}_i^U$ either equals $\mathbf{M}_c$ or satisfies the condition that $\hat{\mathbf{y}}_{i,c}^U = 0$, $0 \le \hat{\mathbf{y}}_{i,k}^U \le 1$, $\forall k \ne c$.*
- *If $\hat{\mathbf{M}}_c^U$ equals $\mathbf{M}_c$, then for any $i \in \{1, \dots, |\hat{\mathcal{G}}^U|\}$, $\hat{\mathbf{y}}_i^U$ is a one-hot vector.*

4.4. Model training

To enable joint training of the encoder and classifier within the CGD-Hyper framework, we design learning objectives leveraging both labeled and unlabeled data. Specifically, we preserve latent node relationships by utilizing the stability of hyperedge homogeneity during representation learning. Furthermore, we employ cross-entropy loss on labeled data for explicit supervision, and enforce consistency between pseudo-label encodings and true label encodings of reliable unlabeled samples for implicit supervision.

To effectively preserve the latent relationships among nodes, we first aggregate the learned node representations to construct hyperedge representations. The calculation is as follows:

$$\widetilde{\mathbf{Z}}_i^e = \frac{1}{|e_i|} \sum_{v_j \in e_i} \widetilde{\mathbf{Z}}_j, \tag{14}$$

where $\widetilde{\mathbf{Z}}_i^e$ denotes the embedding vector of the ith edge e_i, which is computed as the average of the embedding vectors of all nodes contained within the edge. Subsequently, the hyperedge representation is fed into a two-layer Multi-Layer Perceptron $R(\cdot)$ to generate a hyperedge homogeneity vector. To ensure that the representation learning process maintains the inherent characteristics of the graph structure, we utilize the consistency between the hyperedge homogeneity vector originally computed and the one generated post-representation learning as a constraint to guide the node representation learning process.

$$\mathcal{L}_1 = MSE(R(\widetilde{\mathbf{Z}}^e), e_{homo}), \tag{15}$$

where e_{homo} represents the initially computed hyperedge homogeneity vector, where the ith element corresponds to the homogeneity score of the ith hyperedge.

In order to transform the implicit categorical information in unlabeled graphs into quantifiable supervisory signals, we constrain $\hat{\mathbf{M}}^U$ to approximate the ground-truth label encoding $\mathbf{M} \in \mathbb{R}^{C\times C}$, where $\mathbf{M}$ is an identity matrix. The loss function for unlabeled graphs is defined as:

$$\mathcal{L}_2 = D_{KL}(\mathbf{M} \parallel \hat{\mathbf{M}}^U), \tag{16}$$

where D_{KL} denotes the Kullback–Leibler (KL) divergence.

To extract supervision signals from labeled data, we employ the cross-entropy loss function. For each labeled graph in $\mathcal{G}^L$, the classification results are generated by the classifier and supervised using cross-entropy loss, as defined:

$$\mathcal{L}_3 = \frac{1}{|\mathcal{G}^L|} \sum_{i=1}^{|\mathcal{G}^L|} \sum_{j=1}^{c} \mathbf{y}_{ij} \log_2 \hat{\mathbf{y}}_{ij}, \tag{17}$$

where $\mathbf{y}_i$, $\hat{\mathbf{y}}_i$ represent the ground-truth label and predicted probability of ith graph.

The final loss function, which combines both labeled and unlabeled graphs contributions, is given by:

$$\mathcal{L} = \beta_1 \mathcal{L}_1 + \beta_2 \mathcal{L}_2 + \mathcal{L}_3, \tag{18}$$

where $\beta_1, \beta_2 \in [0, 1]$ represent the weights of hyperedge homogeneity loss and unsupervised loss.

To better understand the proposed model, the overall algorithm of CGD-Hyper is listed in Algorithm 1.

Algorithm 1 CGD-Hyper Framework

Input: Labeled graph set $\mathcal{G}^L = \{G_1, \dots, G_{N_L}\}$, unlabeled graph set $\mathcal{G}^U = \{G_{N_L+1}, \dots, G_N\}$, hyperedge Size k, threshold α, loss weight β_1, β_2
Output: Trained model $f : \mathcal{G}^U \to \mathbf{y}$

Dual-Hypergraph Construction:
for each node $v_i \in V$ **do**
  Construct feature hyperedge $\mathbf{H}_i^F$ via top-k similarity
  Construct structure hyperedge $\mathbf{H}_i^S$ via neighbors
  if $\sum_j \mathbf{H}_{ij}^S < \alpha$ **then**
    Augment with feature hyperedges: $\mathbf{H}_j^S \leftarrow \mathbf{H}_j^F$
  end if
end for
Hypergraph Encoding:
Compute hyperedge homogeneity $homo(e)$; Encode to get $\mathbf{Z}^F, \mathbf{Z}^S$
Fuse representations via Co-Gating: $\tilde{\mathbf{Z}} = \mathbf{B} \odot \mathbf{Z}^F + (1 - \mathbf{B}) \odot \mathbf{Z}^S$
Confidence-Aware Label Encoding:
Cluster $\mathcal{G}^U$ to get centers $\mathbf{q}_{c'}^U$; Match with labeled centers $\mathbf{q}_c^L$
Fuse clustering/classification confidences: $y_i^* = \text{argmax}(\mathbf{y}_i \odot \hat{\mathbf{y}}_i)$
Sample reliable graphs $\hat{\mathcal{G}}^U$ and compute label encoding $\hat{\mathbf{M}}^U$
Model Training:
Compute joint loss $\mathcal{L} = \beta_1 \mathcal{L}_1 + \beta_2 \mathcal{L}_2 + \mathcal{L}_3$ and update f
return Trained model f

5. Experiment

In this section, a series of extensive experiments are conducted to systematically evaluate the effectiveness of the proposed CGD-Hyper framework, focusing on the following research questions:

- **RQ1:** How does CGD-Hyper perform in terms of classification accuracy compared to existing baseline methods under data scenarios with scarce labels?
- **RQ2:** How do key components in the CGD-Hyper framework collaborate to enhance overall performance?
- **RQ3:** Can augmenting structural hyperedges with feature hyperedges improve model performance?
- **RQ4:** Is the performance of CGD-Hyper significantly sensitive to the selection of hyperparameters?
- **RQ5:** Can the construction of a dual-hypergraph effectively promote representation learning?

5.1. Experimental setup

In this subsection, we present the datasets used, baseline methods compared and hyper-parameters settings for our experiments.

Table 1
Dataset statistics.

Datasets	Graph Num	Avg.nodes	Avg.edges	classes
PROTEINS	1113	39.06	72.82	2
DD	1178	284.32	715.66	2
COLLAB	5000	74.49	2457.78	3
DBLP	19 456	10.48	19.65	2

5.1.1. Datasets

Building upon previous studies, we evaluate our framework on four widely-used bioinformatics and social network datasets. The specific datasets are: (1) Bioinformatics: **PROTEINS**, **DD**; (2) Social Network: **COLLAB**, **DBLP**; Table 1 provides a detailed description of the statistical information of the datasets.

We adopt the same experimental split as HEAL [18], randomly sampling from the datasets to create training, testing, and validation sets in a ratio of 7:1:2. Within the training set, 1/7(10% of all data) of the graphs are labeled for training purposes.

5.1.2. Baselines

To highlight the advantages of our approach, we compare CGD-Hyper with a set of competitive benchmarks, which are grouped into three categories: (1) **Traditional semi-supervised methods:** Mean-Teacher [40], VAT [41]; (2) **Graph-specific semi-supervised methods:** InfoGraph [22], GraphCL [23], DualGraph [42], HEAL [18]; (3) **Active learning methods:** GALAXY [43], ASGNN [9], GraphSpa [12]; (4) **Pre-training methods:** GPT-GNN [29], Surface-VQMAE [28], MOICLR [30].

5.1.3. Parameter settings

For the implementation of CGD-Hyper, we set the embedding dimension to 32, the batch size to 128, and the training duration to 400 epochs. In the dual-hypergraph construction module, based on experimental results, we initially set the vertex neighbor count k for the feature hypergraph to 20 and the hyperedge size threshold α for the structure hypergraph to 12. However, considering the relatively small number of graph nodes in the employed DBLP dataset, to more effectively construct the hypergraph structure and avoid overfitting, we ultimately adjusted these two parameters to 8 and 6. Additionally, we set the weight hyper-parameters $\beta_1 = 0.3$ and $\beta_2 = 0.4$. The CGD-Hyper model is optimized using the Adam optimizer with an initial learning rate of 0.01 and weight decay set to 0.0005. For the remaining baseline models, their parameter configurations are set according to the specifications in their respective papers. Following previous studies [12,18], we utilize average classification accuracy (ACC) and F1 score as the primary evaluation metrics.

5.2. Performance analysis (RQ1)

We conduct thorough comparisons between CGD-Hyper and baseline methods across all datasets, as detailed in Table 2, to validate its superior performance.

Experimental results show that traditional semi-semi-supervised methods perform the worst due to their inability to capture complex higher-order interaction patterns in graph structures. Pre-training methods outperform traditional ones but still lag behind graph-specific semi-supervised approaches, constrained by the bias of unsupervised signals. Although active learning methods optimize decision boundaries through intelligent sampling, they are less effective than HEAL in leveraging structural information from unlabeled data. CGD-Hyper achieves the best performance on all datasets, primarily due to three advantages: First, compared to HEAL, its dual-hypergraph mechanism extracts higher-order relationships more comprehensively; second, unlike GraphSpa, its confidence-aware strategy more efficiently utilizes unlabeled data to provide supervisory signals; third, in contrast to pre-training methods with separate training, CGD-Hyper enables collaborative optimization of labeled and unlabeled data, avoiding suboptimal representation learning.

To evaluate the robustness of CGD-Hyper under varying proportions of labeled data, we conducted comprehensive experiments on the PROTEINS and IMDB-B datasets, with the results illustrated in Fig. 3.

Experimental results show that all methods improve with more labeled data, confirming the critical role of labels in semi-supervised learning. CGD-Hyper consistently outperforms all baselines across different label proportions, benefiting from its multi-hypergraph mechanism that captures complex high-order semantics and its confidence-aware label encoding that provides reliable supervision.

As labeled data decreases, graph-specific methods (e.g., HEAL) suffer the most significant performance drop, as they heavily rely on labeled data for supervision. Active learning methods, while not optimal at higher label ratios (e.g., 10%), perform well under extreme scarcity (1%) by dynamically selecting the most informative samples to maximize limited labeling budgets. Pre-trained methods are the least sensitive to label proportion changes, since their encoders are pre-trained on large-scale unlabeled data, providing robust feature representations; only the downstream classifier is affected when labels vary.

5.3. Ablation study (RQ2 & RQ3)

To evaluate the contributions of CGD-Hyper's components, we perform a systematic ablation study on the COLLAB dataset using a baseline model that includes a GNN encoder, a classifier, and standard cross-entropy loss. Subsequently, we incrementally introduce key components. The key insights from Table 3 are summarized as follows:

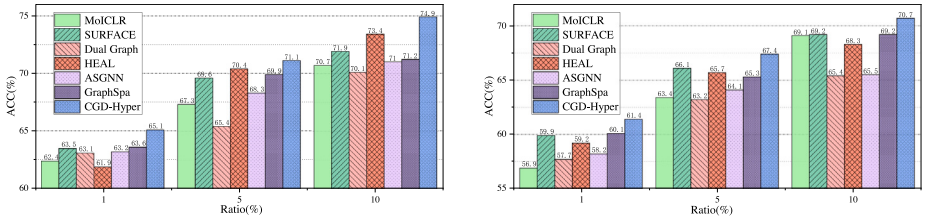
Firstly, the standalone GNN encoder struggles with capturing high-order semantic relationships, whereas the integration of hypergraphs (Hyper-F and Hyper-S) significantly enhances the extraction of meaningful graph representations. Secondly, leveraging multiple high-order relationships leads to improved performance due to their complementary interactions. Thirdly, the introduction of the homophily-aware loss $\mathcal{L}_1$ effectively captures potential dependencies between nodes, thereby enhancing model performance. Finally, the label encoding $\mathcal{L}_2$ module substantially boosts performance by utilizing unlabeled graphs as auxiliary supervision, thereby demonstrating the informational value of unlabeled graphs. Moreover, further optimization through the Confidence-Aware (CA) mechanism indicates that the indiscriminate use of unlabeled graphs is suboptimal, as noisy examples can compromise training stability.

To evaluate the effectiveness of the co-gating method in representation fusion, we conduct a series of comparative experiments. In these trials, we replace the co-gating method with three alternative methods for representation fusion: vector addition (ADD), attention-based fusion (ATT), and multi-layer perceptron (MLP) fusion. Our evaluations are carried out on the PROTEINS and DD datasets, with the results displayed in Fig. 4. Clearly, the CG achieves substantial performance enhancements over the other fusion methods across both datasets. This robustly indicates the efficacy of the co-gating method in merging hypergraph representations from dual perspectives. Furthermore, both ATT and MLP fusion outperform the ADD method, underscoring the necessity for domain-specific inductive biases in feature fusion. This highlights that simplistic arithmetic operations are insufficient for meeting the intricate demands of complex graph learning tasks.

To investigate the effectiveness of augmenting structural hyperedges with feature hyperedges, we conduct ablation studies across four datasets to compare the model performance with (W/) and without (W/O) this augmentation strategy. As illustrated in Fig. 5, the results consistently demonstrate that incorporating feature hyperedges yields superior performance. Notably, the improvements are particularly pronounced on the PROTEINS and DD datasets. This can be attributed

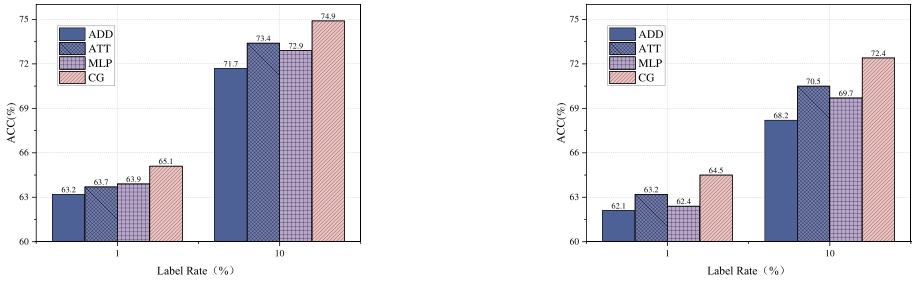
Table 2
The average accuracy (ACC) and F1 score results of semi-supervised graph classification across four datasets. The **optimal performance** is highlighted in bold, while the suboptimal performance is underlined.

Datasets	PROTEINS		DD		COLLAB		DBLP	
Models	ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)
Mean-Teacher	64.3 ± 2.1	62.9 ± 1.7	60.6 ± 1.8	59.7 ± 2.1	63.6 ± 1.4	63.5 ± 1.3	64.6 ± 0.4	63.3 ± 1.2
VAT	64.1 ± 1.2	61.7 ± 2.2	59.9 ± 2.6	59.5 ± 2.4	64.1 ± 1.1	62.3 ± 0.9	63.7 ± 0.5	63.1 ± 0.8
InfoGraph	68.2 ± 0.7	64.3 ± 1.5	67.5 ± 1.4	65.9 ± 1.2	65.7 ± 0.4	62.9 ± 0.7	63.7 ± 0.8	61.2 ± 1.1
GraphCL	69.4 ± 0.8	67.9 ± 1.3	68.7 ± 1.2	65.9 ± 1.7	66.4 ± 0.6	64.7 ± 0.6	65.1 ± 0.9	64.1 ± 1.5
Dualgraph	70.1 ± 1.2	67.5 ± 1.0	69.8 ± 0.8	68.6 ± 1.4	67.2 ± 0.6	62.3 ± 1.5	65.4 ± 1.1	64.3 ± 1.3
HEAL	73.4 ± 0.8	71.8 ± 1.4	72.1 ± 0.9	68.4 ± 1.7	68.3 ± 0.5	68.1 ± 0.6	68.2 ± 1.4	65.9 ± 1.2
GALAXY	70.2 ± 0.5	68.6 ± 0.8	71.2 ± 0.7	69.7 ± 1.2	64.4 ± 0.8	62.6 ± 0.8	66.1 ± 1.3	64.8 ± 1.7
ASGNN	71.0 ± 0.7	70.2 ± 1.1	71.1 ± 0.9	70.5 ± 0.8	65.5 ± 1.1	61.2 ± 1.6	64.3 ± 0.5	63.4 ± 0.9
GraphSpa	71.2 ± 0.7	70.8 ± 0.9	71.4 ± 0.8	70.7 ± 1.4	66.9 ± 0.9	65.8 ± 0.7	69.7 ± 1.1	68.4 ± 1.4
GPT-GNN	64.5 ± 1.1	61.3 ± 1.2	62.2 ± 0.7	61.8 ± 0.5	63.2 ± 1.1	62.5 ± 1.4	67.4 ± 1.7	66.2 ± 2.4
MOICLR	70.7 ± 2.1	68.4 ± 1.6	67.4 ± 1.4	62.3 ± 1.1	68.5 ± 0.8	66.1 ± 1.8	69.1 ± 1.3	67.5 ± 1.1
Surface-VQMAE	71.9 ± 0.7	70.5 ± 0.8	72.6 ± 0.6	72.3 ± 0.7	67.7 ± 1.2	67.3 ± 1.0	69.2 ± 0.4	68.4 ± 0.6
CGD-Hyper	**74.9 ± 0.6**	**73.6 ± 0.8**	**74.5 ± 0.7**	**73.8 ± 0.6**	**70.7 ± 1.3**	**70.1 ± 0.9**	**72.4 ± 0.8**	**71.5 ± 1.2**



(a) PROTEINS. (b) COLLAB.

Fig. 3. Comparison of CGD-Hyper with baselines by varying the labeling ratio of training graphs on the PROTEINS and COLLAB Datasets. The proportions of labeled graphs in the total are 1%, 5%, and 10%, respectively.



(a) PROTEINS. (b) DBLP.

Fig. 4. The experimental results of replacing the co-gating mechanism with different fusion mechanisms on PROTEINS and DBLP.

to the prevalence of isolated and low-connectivity nodes within these datasets; the proposed augmentation strategy effectively alleviates connectivity limitations, thereby facilitating high-order message passing and uncovering richer structural semantics.

5.4. Hyper-parameters analysis (RQ4)

This section explores the effects of key hyper-parameters, including the hyperedge size k in the feature hypergraph, the threshold α in the structure hypergraph, and the loss weight β_1, β_2.

5.4.1. Impact of hyperedge size k and threshold α

We evaluate the impact of hypergraph construction parameters k and α on model ACC across the range $\{4, 8, \ldots, 32\}$ on the DD and DBLP datasets. As illustrated in Fig. 6, the results reveal that model performance exhibits significant sensitivity to parameter selection. Specifically, optimal performance is achieved at $k = 20$ and $\alpha = 12$ on the DD dataset, while the DBLP dataset peaks at $k = 8$ and $\alpha = 6$. Further observation indicates that performance degrades as the parameter values deviate from these optima. This phenomenon suggests that appropriate parameter settings are critical for effectively capturing the high-order relationships inherent in the data: values that are too small may lead to information loss, whereas values that are too large may introduce redundant connections, thereby weakening the model's discriminative power.

Table 3
Results of ablation experiments conducted on the COLLAB dataset.

Model components	Hyper-F	Hyper-S	$\mathcal{L}_1$	$\mathcal{L}_2$	CA	Performance
Base model						59.8
	✓					62.2
		✓				61.7
	✓		✓			62.9
	✓	✓				65.1
	✓	✓	✓			65.6
	✓	✓	✓	✓		69.8
	✓	✓	✓	✓	✓	70.7

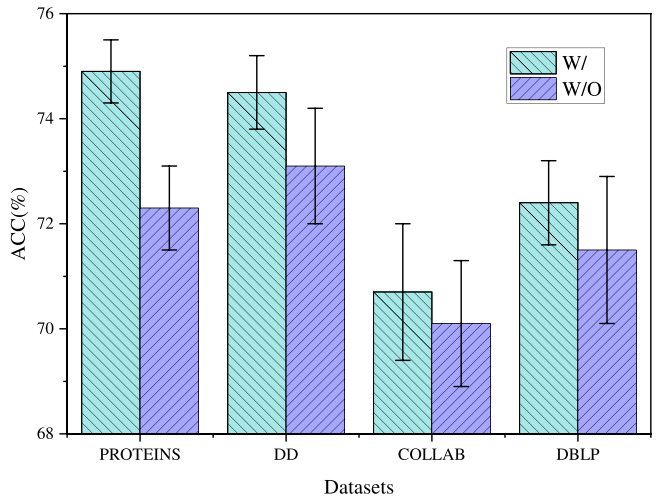


Fig. 5. Comparison of model performance with (W/) and without (W/O) feature hyperedge augmentation across four datasets.

Furthermore, we investigate the impact of hyperedge size k and structure threshold α on homogeneity. As shown in Fig. 7, on the PROTEINS, DD, and COLLAB datasets, the homogeneity of feature hypergraphs peaks at $k = 20$ and declines as k deviates from this value. In contrast, the optimal k for the DBLP dataset is 8. With the optimal k fixed, we further examined the effect of structural hyperedges. The results indicate that, except for DBLP, which achieves optimal homogeneity at $\alpha = 6$, all other datasets reach their maximum homogeneity at $\alpha = 12$. These findings suggest that hyperedge homogeneity is highly sensitive to both hyperparameters.

5.4.2. *Influence of loss weight β_1 and β_2*

In Fig. 8, we evaluate how the semi-supervised loss weights $\beta_1, \beta_2 \in [0.1, 1]$ affect classification accuracy across three benchmark datasets: DD and COLLAB.

As shown in Fig. 8, the model achieves optimal performance when $\beta_1 = 0.3$ and $\beta_2 = 0.4$, while other parameter combinations lead to a monotonic performance decline. The reasons can be summarized as follows: Firstly, when $\beta_1 < 0.3$, the model fails to adequately capture node correlations within hyperedges, resulting in information loss that limits performance improvement. Secondly, when $\beta_1 > 0.3$, performance degradation is attributed to over-reliance on potentially noisy dependencies in the initial hyperedges, which compromises model generalization. Furthermore, when $\beta_2 < 0.4$, the model insufficiently exploits latent information in unlabeled graphs, leading to weak supervision signals and suboptimal performance. Finally, when $\beta_2 > 0.4$, further performance decline indicates that overemphasizing unlabeled graphs undermines the dominant role of labeled graphs as reliable supervision sources in semi-supervised learning, adversely affecting the training process.

5.5. *Case study (RQ5)*

To showcase the effectiveness of representation extraction from the dual-hypergraph, we perform average pooling on the representations before and after fusion on the DD dataset, and employ t-SNE for visualization. Two fusion methods, ADD and CG, are adopted for comparison. As illustrated in Fig. 9, the embedding distributions of the feature hypergraph $\hat{\mathbf{Z}}^F$ and the structure hypergraph $\hat{\mathbf{Z}}^S$ exhibit significant disparities, confirming the complementary nature of their information. However, the linear fusion via the ADD method leads to degradation in the feature space, characterized by blurred class boundaries (as highlighted by the dashed ellipses in Fig. 9(c)) and increased intra-class dispersion. Consequently, its Silhouette Score (0.22) is notably lower than that of the pre-fusion $\hat{\mathbf{Z}}^S$ (0.29). In contrast, the CG method adeptly merges multi-view features through its dynamic weighting mechanism. It not only demonstrates enhanced intra-class compactness and inter-class separation (as shown in Fig. 9(d)) but also achieves the highest Silhouette Score of 0.37, significantly outperforming the other representations. This strongly underscores the proficiency of CG in feature selection and noise mitigation, highlighting its substantial potential in multi-view fusion tasks.

6. Conclusion

In this work, we address the critical challenge of label scarcity in semi-supervised graph classification by introducing the CGD-Hyper model. Our framework effectively captures high-order semantic relationships by concurrently integrating node attributes and structural topology through a dual hypergraph mechanism, fused via an adaptive co-gating method. Furthermore, the proposed confidence-aware label encoding strategy successfully mitigates the noise issues inherent in pseudo-labeling by jointly leveraging clustering and classification confidences. Extensive experiments on four benchmark datasets demonstrate the superiority of CGD-Hyper, achieving an average accuracy improvement of 3.5% over state-of-the-art methods.

Despite the promising results, CGD-Hyper has certain limitations. First, the construction of hypergraphs introduces additional computational overhead compared to standard GNNs, which may pose challenges for extremely large-scale graphs. Future work could explore more efficient hypergraph construction strategies to improve scalability. Second, the setting of hyperedge size, while demonstrating robustness in our experiments, is currently determined empirically. Future research could investigate differentiable or adaptive mechanisms to dynamically learn the optimal hyperedge size, thereby enhancing the model's adaptability across diverse datasets.

CRediT authorship contribution statement

Hui Shen: Visualization, Validation, Methodology. **Huifang Ma:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis. **Rui Bing:** Writing – review & editing, Supervision. **Jiyuan Sun:** Visualization, Data curation. **Zhixin Li:** Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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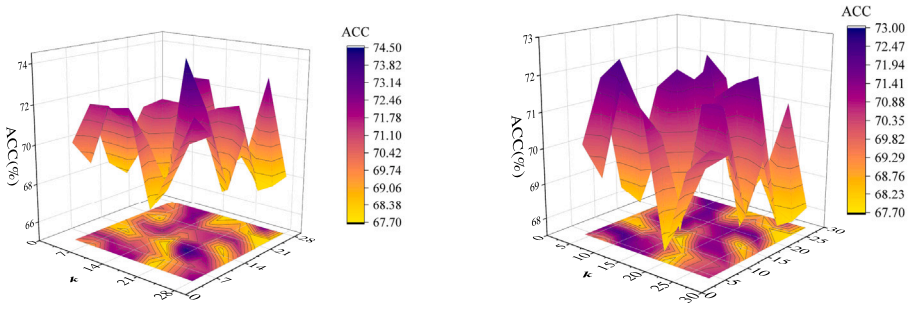
Appendix. Proof for Theorem 4.1

(1) The sum of all elements in $\hat{\mathbf{M}}_c^U$, i.e., $1^T\hat{\mathbf{M}}_c^U$, can be calculated as follows:

$$1^T\hat{\mathbf{M}}_c^U = \frac{1^T\sum_{i=1}^{|\hat{\mathcal{G}}^U|}(\hat{\mathbf{y}}_i^U\hat{\mathbf{y}}_i^U)}{\sum_{i=1}^{|\hat{\mathcal{G}}^U|}\hat{\mathbf{y}}_{i,c}^U} = \frac{\sum_{i=1}^{|\hat{\mathcal{G}}^U|}(\hat{\mathbf{y}}_{i,c}^U(1^T\hat{\mathbf{y}}_i^U))}{\sum_{i=1}^{|\hat{\mathcal{G}}^U|}\hat{\mathbf{y}}_{i,c}^U} = \frac{\sum_{i=1}^{|\hat{\mathcal{G}}^U|}\hat{\mathbf{y}}_{i,c}^U}{\sum_{i=1}^{|\hat{\mathcal{G}}^U|}\hat{\mathbf{y}}_{i,c}^U} = 1. \quad \text{(A.1)}$$

(2) Since $0 \le \hat{\mathbf{y}}_{i,c}^U \le 1, \forall j \in \{1, \dots, C\}, \forall i \in \{1, \dots, |\hat{\mathcal{G}}^U|\}$, based on Eq. (13) and the Property (1) in Theorem 4.1, we have:

$$0 \le \hat{\mathbf{M}}_{c,j}^U \le 1, \forall j \in \{1, \dots, C\}. \quad \text{(A.2)}$$



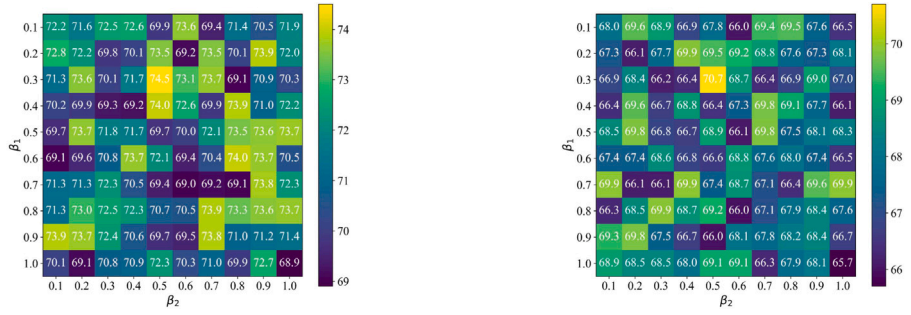
(a) DD (b) DBLP

Fig. 6. Performance of CGD-Hyper on DD and DBLP with various hyperedge size k and threshold α.



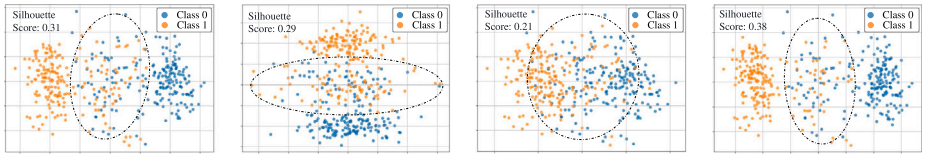
(a) Feature hyperedge homophily (b) Structural hyperedge homophily

Fig. 7. Analysis of the impact of feature hyperedge size and structural threshold on homophily. (a) The variation trends of feature hypergraph homophily under different hyperedge sizes k. (b) The impact of structural threshold α on structural hypergraph homophily with the optimal hyperedge size fixed.



(a) DD (b) COLLAB

Fig. 8. Investigation of loss weight β_1 and β_2. The loss weight β_1 and β_2 is set within the range $[0.1, 1.0]$, and the model is deployed on the DD and COLLAB datasets.



(a) t-SNE of $\hat{\mathbf{Z}}^F$ (b) t-SNE of $\hat{\mathbf{Z}}^S$ (c) t-SNE on ADD (d) t-SNE on CG

Fig. 9. The visualization of learned graph representations with average pooling through t-SNE technology: including $\hat{\mathbf{Z}}^F$ and $\hat{\mathbf{Z}}^S$ before fusion, as well as $\hat{\mathbf{Z}}$ after fusion via the ADD and CG methods.

(3) Since $\hat{\mathbf{y}}_i^U$ is the ground-truth label encoding of sample $\mathbf{x}_i^u$, $\hat{\mathbf{y}}_{i,c}^U = 1$ if $\mathbf{x}_i^u$ belongs to category c, else $\hat{\mathbf{y}}_{i,c}^U = 0$. Accordingly, we can calculate the cth element of $\hat{\mathbf{M}}_c^U$ as follows:

$$\hat{\mathbf{M}}_{c,c}^U = \frac{\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U \hat{\mathbf{y}}_{i,c}^U}{\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U} = \frac{|\hat{\mathcal{G}}^U|_c}{|\hat{\mathcal{G}}^U|_c} = 1, \tag{A.3}$$

where $|\hat{\mathcal{G}}^U|_c$ is the number of unlabeled samples belonging to category c. Similarly, the kth ($\forall k \neq c$) element of $\hat{\mathbf{M}}_c^U$ can be calculated as follows:

$$\hat{\mathbf{M}}_{c,k}^U = \frac{\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U \hat{\mathbf{y}}_{i,k}^U}{\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U} = \frac{0}{|\hat{\mathcal{G}}^U|_c} = 0, \forall k \neq c. \tag{A.4}$$

Hence, $\hat{\mathbf{M}}_c^U$ equals $\mathbf{M}_c$.

$$\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U \hat{\mathbf{y}}_{i,k}^U = 0. \tag{A.5}$$

Since $\hat{\mathbf{y}}_{i,c}^U \hat{\mathbf{y}}_{i,k}^U \geq 0$, we have

$$\hat{\mathbf{y}}_{i,c}^U \hat{\mathbf{y}}_{i,k}^U = 0, \quad \forall k \neq c. \tag{A.6}$$

When $\hat{\mathbf{y}}_{i,c}^U > 0$, we have $\hat{\mathbf{y}}_{i,k}^U = 0, \forall k \neq c$, which implies $\hat{\mathbf{y}}_i^U = \mathbf{M}_c$. When $\hat{\mathbf{y}}_{i,c}^U$ equals 0, Eq. (A.6) naturally holds.

By combining the above two cases, Eq. (A.5) is equivalent to that for some $c \in \{1, \ldots, C\}$ and any $i \in \{1, \ldots, |\hat{\mathcal{G}}^U|\}$,

$$\hat{\mathbf{y}}_{i,c}^U = 1, \hat{\mathbf{y}}_{i,k}^U = 0, \forall k \neq c \Rightarrow \hat{\mathbf{y}}_i^U = \mathbf{M}_c \quad \text{or} \quad \hat{\mathbf{y}}_{i,c}^U = 0, 0 \leq \hat{\mathbf{y}}_{i,k}^U \leq 1, \forall k \neq c. \tag{A.7}$$

Moreover, since $\hat{\mathbf{M}}_{c,c}^U = 1$, we obtain

$$\hat{\mathbf{M}}_{c,c}^U = \frac{\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U \hat{\mathbf{y}}_{i,c}^U}{\sum_{i=1}^{|\hat{\mathcal{G}}^U|} \hat{\mathbf{y}}_{i,c}^U} = 1 \Rightarrow \sum_{i=1}^{|\hat{\mathcal{G}}^U|} ((\hat{\mathbf{y}}_{i,c}^U)^2 - \hat{\mathbf{y}}_{i,c}^U) = 0. \tag{A.8}$$

Since $0 \leq \hat{\mathbf{y}}_{i,c}^U \leq 1, \forall i \in \{1, \ldots, |\hat{\mathcal{G}}^U|\}$, we can get $(\hat{\mathbf{y}}_{i,c}^U)^2 \leq \hat{\mathbf{y}}_{i,c}^U$. To make Eq. (A.8) hold, we must have $(\hat{\mathbf{y}}_{i,c}^U)^2 = \hat{\mathbf{y}}_{i,c}^U$, which implies that

$$\hat{\mathbf{y}}_{i,c}^U = 1 \quad \text{or} \quad \hat{\mathbf{y}}_{i,c}^U = 0, \forall i \in \{1, \ldots, |\hat{\mathcal{G}}^U|\}, \tag{A.9}$$

which is consistent with Eq. (A.7).

(5) Since $\hat{\mathbf{M}}_c^U$ equals $\mathbf{M}_c$ for any $c \in \{1, \ldots, C\}$, based on Eq. (A.9), we have

$$\hat{\mathbf{y}}_{i,c}^U = 1 \quad \text{or} \quad \hat{\mathbf{y}}_{i,c}^U = 0, \quad \forall i \in \{1, \ldots, |\hat{\mathcal{G}}^U|\}, \forall c \in \{1, \ldots, C\}. \tag{A.10}$$

Hence each element in $\hat{\mathbf{y}}_i^U$ is either 0 or 1, and because $\hat{\mathbf{y}}_i^U$ denotes the predicted probabilities for C categories, it is evident that $\hat{\mathbf{y}}_i^U$ is a one-hot vector for any $i \in \{1, \ldots, |\hat{\mathcal{G}}^U|\}$.

Data availability

The data used are all publicly available for research by previous researchers.

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